Project Design Phase-II
Data Flow Diagram & User Stories

Date	21 October 2026
Team ID	EN25CS3010849
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	4 Marks

Data Flow Diagrams:

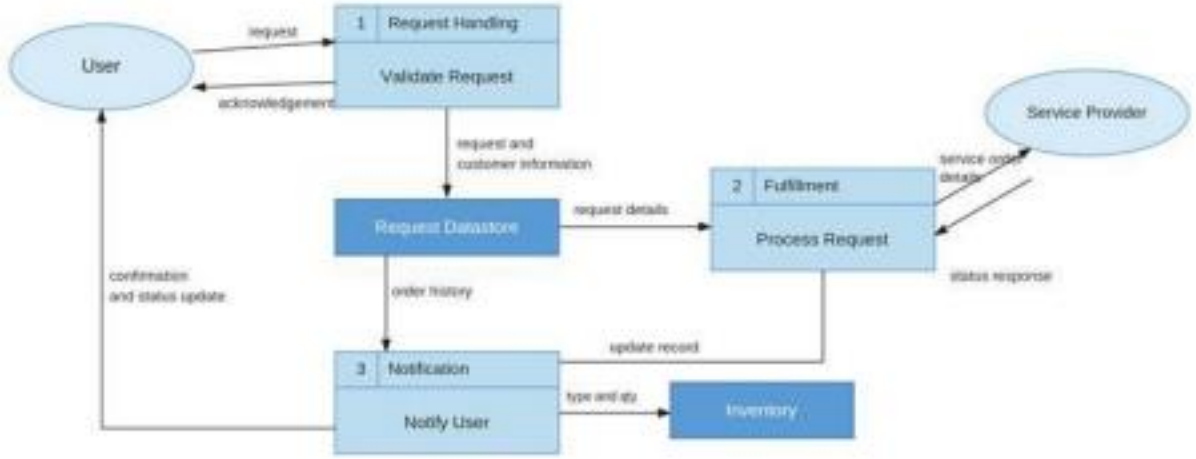
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	**User / Sports Analyst:** Sends query requests (e.g., filter by year, country, or sport) and views interactive dashboard visualizations. **Olympic Data Source / CSV Repositories:** External data source providing raw datasets (medals.csv, athlete_events.csv, athlete.csv).
Rectangle with numbered header	**Process**	**1.0 Data Ingestion & Preprocessing:** Cleans raw CSV files, handles missing values, removes duplicates, and standardizes text formats. **2.0 Metrics & Aggregation Processing:** Computes total medal tallies, calculates ratios (e.g., success rates, physical attribute correlations), and processes custom dynamic queries. **3.0 Dashboard Visualization Rendering:** Generates interactive visual charts, bar plots, scatter plots, and KPI cards for the analyst.

Rectangle (solid fill, no number)	**Data Store**	**D1: Cleaned Olympic Database / Data Store:** Stores processed athlete details, sports metadata, and aggregated country medal tallies.
Labeled arrow	**Data Flow**	**Raw Datasets (medals.csv, athlete.csv):** Flows from *Olympic Data Source → 1.0 Data Ingestion*. **Cleaned Datasets:** Flows from *1.0 Data Ingestion → D1 Data Store*. **Filter & Query Request:** Flows from *User / Sports Analyst → 2.0 Metrics Processing*. **Processed Aggregates:** Flows from *D1 Data Store → 2.0 Metrics Processing*. **Visualization Data Stream:** Flows from *2.0 Metrics Processing → 3.0 Dashboard Rendering*. **Interactive Dashboard / Graphs:** Flows from *3.0 Dashboard Rendering → User / Sports Analyst*.

Example: DFD Level 0 (Industry Standard)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile user)	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	I can access my account / dashboard	High	Sprint-1
		USN-2	As a user, I will receive confirmation email once I have registered for the application	I can receive confirmation email & click confirm	High	Sprint-1
		USN-3	As a user, I can register for the application through Facebook	I can register & access the dashboard with Facebook Login	Low	Sprint-2
		USN-4	As a user, I can register for the application through Gmail		Medium	Sprint-1
	Login	USN-5	As a user, I can log into the application by entering email & password		High	Sprint-1
	Dashboard					
Customer (Web user)						
Customer Care Executive						
Administrator						